WONDER DAILY

How Speakers Turn Waves Into Sound

A wiggle begins

Little heard a note from the speaker. A wiggly electric signal made something move. Could the notebook's smooth wave become a sound?

Push, then pull

A wire coil sat near a magnet. The changing signal made it push and pull. The attached paper cone moved back and forth.

A tiny hop

Little hummed near the rice bowl. Big gently tapped its side. The tight wrap vibrated, and rice hopped. A small motion became visible.

Two kinds of change

Rice hopped on the vibrating drumhead too. Faster wiggles make higher sounds. Bigger wiggles make louder sounds. Little kept the taps gentle.

Along a string

Big spoke into a paper cup. Its bottom vibrated. A taut string carried the motion to Little's cup, which moved the air nearby.

A wiggle reaches you

The cone sent pressure waves through air. Little's eardrum vibrated when they arrived. Where would the wiggle travel next?

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!